

BFU10: Database Development

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SECTION 1: PURPOSE AND CLIENT REQUIREMENTS

Purpose:

The purpose is to create a digital database system for a surgery, it will contain patient and appointment data. They want to move from a paper-based system they are currently using, as there is a planned reorganization of the health services in the area, which will increase the number of patients and doctors, which has made management decide to upgrade their systems. The surgery is going to be the pilot for the local health authority to get the rest of the organization to upgrade to the digital database systems and move away from paper based.

Clients Requirements:

The client's requirements are that we don't do anything that exceeds their budget, and that it should be relational database, there should be 2 tables, one with the patients details and another with appointment details. We must import the existing spreadsheet of data into the tables, queries must also be made, some of these are a list of patients all in the DE1 area, with their name, address, sorted alphabetically by surname. Another query is a list of new patients in the last 6 months to monitor the growth of the surgery, it must list their name, registration date, and their doctor's name. and sorted by date. Another one is a list of patients over 50 years old to be contacted for their annual flu jab, it must list their name, address, town and postcode and be sorted alphabetically by their Last Name ascending. Finally, another query is for a list of under 5 patients to check if their immunizations are up to date. The client wants the database to be in a professional setting, using consistent themes and displays their health authority logo.

SECTION 2: HARDWARE AND SOFTWARE NEEDS

There is a very limited budget for hardware and software, including training and development. We must choose software that can run on possibly existing hardware. The software we choose should also be easy to use for the average person, as there aren't enough resources available to teach staff on how to use the software. Because of this I've decided to choose Microsoft Access, as it's easy to setup, maintain and for the average user

it's not a steep learning curve. We can create more human readable formats and menus for ease of navigation. The systems within the surgery probably will not need hardware upgrades, as the system requirements for Microsoft Access are very low, requiring only a 1.6GHz dual core CPU, and 4GB of RAM for 64-bit, and 2GB for 32-Bit, and it only needs 4GB of available storage, and at least Windows 10, however there may be LTSC versions of Access if the systems cannot upgrade to 10 or higher, which means it would still get security updates to be remain safe, however you wouldn't see any new features.

SECTION 3: THE DESIGN

A: AT LEAST TWO TABLES WITH APPROPRIATE ATTRIBUTES

PatientRef
FirstName
LastName
Address

Town
Postcode
Date of Birth
Contact number
Date registered

Table 1 Patients
Table 2 Appointments

AppointmentRef
PatientRef
AppointmentDate
AppointmentTime
DoctorsName
Attended

B: DATA DICTIONARY

Table 1 Patients

Field name	Data-type	Field length	Key?	Format	Default	Validation rule and Validation Text
Patient Ref	Short Text	5	Yes	L9999	P	“L9999” - Please only enter/edit valid patient references.
First Name	Short Text	40	No	@	N/A	Len([First Name])>=2 - Please enter a valid first name with a length of 2 to 40 characters.
Last Name	Short Text	40	No	@	N/A	Len([Last Name])>=2 - Please enter a valid last name with a length of 2 to 40 characters.
Address	Short Text	100	No	@	N/A	Len([Address])>=5 - Please enter a valid address.

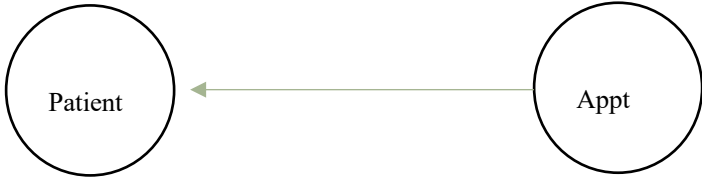
Town	Short Text	85	No	@	N/A	Len([Town])>=1 - Please enter a valid town name, from 1 to 85 characters.
Postcode	Short Text	8	No	@	N/A	"LL09 0LL" - Please input a valid postcode.
Date of Birth	Date/Time	?	No	dd/mm/yyyy;@	N/A	00/00/0000;0;- - Please input a valid date of birth.
Contact Number	Short Text	11	No	00000000000	N/A	Len([Contact Number])<11 - Please enter a valid number, it must be 11 numbers.
Date Registered	Date/Time	?	No	dd/mm/yyyy	N/A	00/00/0000;0;- - Please input a valid date of registration.

Table 2

Field name	Data-type	Field length	Key?	Format	Default	Validation rule and Validation Text
AppointmentRef	Short Text	5	Yes	L9999	A	"L9999" - Please only enter/edit valid appointment references.

PatientRef	Short Text	5	No	L9999	P	“L9999” - Please only enter/edit valid patient references.
AppointmentDate	Date/Time	?	No	dd/mm/yyyy;@	N/A	00/00/0000;0;_ - Please input a valid appointment date.
AppointmentTime	Date/Time	?	No		N/A	Placeholder – Please input a valid time.
DoctorsName	Short Text	80	No	@	N/A	Len([Doctors Name])>=5 - Please enter a valid name with a length of 5 to 80 characters.
Attended	Short Text	1	No	“Y” OR “N”	N/A	Placeholder – Please only enter Y or N.

C: ENTITY DIAGRAM



PatientRef
FirstName
LastName
Address
Town
Postcode
Date of birth
Contact number
Date registered

AppointmentRef
PatientRef
AppointmentDate
AppointmentTime
DoctorsName
Attended



Design 1 – Patients data entry sub-form

PLACEHOLDER

Patient Reference

First Name

Last Name

Address Line

Town

Postcode

Date of Birth

Contact Number

Date Registered

Appointment Ref

Patient Reference

Appointment Date

Appointment Time

Doctors Name

Attended

Cannot edit, as Access will handle this for us.

Validation rules with messages to tell the user to correctly input the data, and guide on what to input.

Calendar button to open a visual menu to select date easily and without confusion on format.

Create new patient data (moves to end, making the form empty, ready for input)

Navigation buttons to navigate through patient data

Back to main menu

Patient appointments listed

Font Style:- Tahoma, Arial

Font Size:- 12, 10

Font Colour:- White, Black

Design 2 – Appointments data entry sub-form

PLACEHOLDER

Appointment Ref

Patient Reference

Appointment Date 

Appointment Time 

Doctors Name

Attended

Cannot edit, as Access handles this for us

Calendar button to open a visual menu to select date easily and without confusion on format.

Can only be Y or N inputted, keeping it simple to store.

Back to main menu

Navigation buttons to navigate through appointment data

Create new appointment (moves to end, making the form empty, ready for input)

Font Style:- Tahoma, Arial

Font Size:- 12, 10

Font Colour:- Black White

Design – Main Menu Form

PLACEHOLDER



Forms

Patient Details Forms

Appt Details Forms

Queries

DE1 Patients moved to Health Centre

DE1 Patients missed Appts in last month

Patients under 5

New Patients in last 6 months

Patients over 50yrs

Reports

Report of DE1 Patients moved to Health Centre

Report of DE1 Patients missed Appts in last month

Report of Patients under 5

Report of new Patients in last 6 months

Report of Patients over 50yrs

Font Style:- Tahoma
Font Size:- 22, 11, 7
Font Colour:- Black, White

F: TEST PLAN

Object	Test	Purpose of test
Main Menu Form	Does it launch upon opening the database?	Simplicity – Easy for user to start, as they don't have to find the form manually.
Main Menu Form	Checking all the buttons for forms, queries and reports go to the right places.	Usability – To make sure that the main menu works as intended for use.
Main Menu Form	Does the close button close the database?	Usability – To be able to close the database when no longer needed.
Patient Table	Is the Patient Ref validated? Is inputting invalid data blocked?	Functionality of the table: checking if validation is working correctly.
Patient Table	Can we enter a valid first name into its designated record and does it save without problems? Is invalid data blocked?	Functionality of the table

Patient Table	Can we enter a valid last name into its designated record and does it save without problems? Is invalid data blocked?	Functionality of the table
Patient Table	Can we enter a valid date of birth into its designated record and does it save without problems? Are invalid dates blocked?	Functionality of the table
Patient Table	Can we enter a valid contact number into its designated record and does it save without problems? Are invalid contact numbers blocked?	Functionality of the table
Patient Form	If you don't enter a contact number fully, is it rejected as intended?	Usability and validation
Patient Form	If you include an @ in the address, does it reject the input? (It is a physical address, not an email)	Usability and validation

Patient Form	Does the next record button work as intended? Does it move to the next record?	Usability of form
Patient Form	Does the previous record button work as intended? Does it move to the previous record?	Usability of form
Patient Form	Does the new record button work as intended? Does it go to an empty record so the user can create the new record?	Usability of form
Patient Form	Can we only input valid postcodes? Are invalid postcodes blocked? Does the validation work properly for correct postcodes?	Usability and validation
Appointments Form	Can we only input Y or N for attending? Are other characters rejected?	Usability and validation
Appointments Form	Does the next record button work as intended? Does it move to the next record?	Usability of form

Appointments Form	Does the previous record button work as intended? Does it move to the previous record?	Usability of form
Appointments Form	Does the new record button work as intended? Does it go to an empty record so the user can create the new record?	Usability of form

G: DATA SET

Additional Excel file. Update to ensure you have 50 records. – Submit with your assignment.

Submitted with assignment.

SECTION 4: QUERIES

1. **Description & Purpose of the Query**

List patients in the DE1 area to be informed they have been transferred to the local health center, displaying their First Name, Last Name, Address, Town and Postcode, which will all be sorted alphabetically by postcode.

Query:

Field	FirstName	LastName	Address	Town	Postcode
Table	Patients	Patients	Patients	Patients	Patients
Sort Ascending, Descending, Not sorted					Ascending
Show? (tick)	/	/	/	/	/
Criteria?					Like "DE1*"

2. Description & Purpose of the Query

List patients in DE1 area that have missed appointments in the last month, displaying First Name, Last Name, Contact Number and Appointment Date, all sorted alphabetically by Last Name.

Query:

Field	FirstName	LastName	ContactNumber	AppointmentDate	Attended	Postcode
Table	Patients	Patients	Patients	Appointments	Appointments	Patients
Sort Ascending, Descending, Not sorted		Ascending				

Show? (tick)	/	/	/	/	☐	☐
Criteria?				>Date()-31	“N”	Like “DE1*”

3. Description & Purpose of the Query

Listing new patients within the last 6 months to monitor the growth of the surgery incase they need more doctors. It must display the First Name, Last Name, Date Registered, Doctors Name, and be sorted descending by date.

Query:

Field	FirstName	LastName	DateRegistered	DoctorsName
Table	Patients	Patients	Patients	Patients
Sort Ascending, Descending, Not sorted			Descending	
Show? (tick)	/	/	/	/
Criteria?			>Date()-183	

4. Description & Purpose of the Query

Listing all patients over 50 that must be contacted for their annual flu jab, it must display the First Name, Last Name, Address, Town, and Postcode, sorted by Last Name in ascending order.

Query:

Field	FirstName	LastName	Address	Town	Postcode	DateOfBirth
Table	Patients	Patients	Patients	Patients	Patients	Patients
Sort Ascending, Descending, Not sorted		Ascending				
Show? (tick)	/	/	/	/	/	☐
Criteria?						<DateAdd("yyyy", -50, Date())

5. Description & Purpose of the Query

Listing patients under the age of 5 to ensure their vaccinations are up to date, the query must display the First Name, Last Name, Address, Town, and Postcode, sorted by Last Name in descending order.

Query:

Field	FirstName	LastName	Address	Town	Postcode	DateOfBirth
Table	Patients	Patients	Patients	Patients	Patients	Patients
Sort Ascending, Descending, Not sorted		Descending				
Show? (tick)	/	/	/	/	/	☐
Criteria?						<DateAdd("yyyy", -5, Date())

Description & Purpose of Report 1-

This will be a report of DE1 Patients that have moved to Health Centre, its purpose is to show a neat list of DE1 patients that need to be informed that they have been transferred to the Health Centre, which would contain their name, address, town and postcode, which would be sorted alphabetically by postcode.

Design

[illegible]

Font Style:- Tahoma, Times New Roman

Font Size:- 24, 14, 10

Font Colour:- White

Description & Purpose of Report 2-

This will be a report that displays all DE1 patients that have missed appointments within the last month, it would display First Name, Last Name, contact number, and appointment date, sorted alphabetically by surname.

Design

Missed DE1 Appts in last month				PLACEHOLDER
First Name	Last Name	Contact Num	Appt Date	

Font Style:- Tahoma, Times New Roman

Font Size:- 24, 14, 10
Font Colour:- White

Description & Purpose of Report 3-

This will be a report that lists all new patients within the last six months so they can easily visualize and share the growth of the surgery, so they can monitor whether they need more doctors, it would need to list First Name, Last Name, Date Registered, and their Doctors Name, and be sorted in descending order by date.

Design

New Patients in last 6 months				PLACEHOLDER
First Name	Last Name	Date Regd	Doctor Name	

Font Style:- Tahoma, Times New Roman
Font Size:- 24, 14, 10
Font Colour:- White

SECTION 6: ALTERNATIVE DESIGNS

Data entry sub-form alternative design – Patients table



Patient Form



1





PLACEHOLDER

Patient Reference

First Name

Last Name

Address Line

Town

Postcode

Date of Birth

Contact Number

Date Registered



Appt Form



1





PLACEHOLDER

Appt Reference

Patient Reference

Appt Date

Appt Time

Doctors Name

Attended

Font Style:- Bahnschrift, Arial
Font Size:- 36, 11, 10
Font Colour:- White, Black

Data entry sub-form alternative design – Appointment table



Appt Form



1





PLACEHOLDER

Appt Reference	
Patient Reference	
Appt Date	
Appt Time	
Doctors Name	
Attended	Y/N

Font Style:- Bahnschrift, Arial

Font Size:- 36, 11, 10

Font Colour:- White, Black

Main menu form alternative design



Main Menu

PLACEHOLDER

Forms

Patient Details
Forms

Appointment Details
Forms

Queries

DE1 Patients moved
to Health Centre

DE1 Patients missed
Appts in last month

New Patients in last
6 months

Patients over 50yrs

Patients under 5

Reports

Report of DE1
Patients moved to
Health Centre

Report of DE1
Patients missed
Appts in last month

Report of new
Patients in last 6
months

Report of Patients
over 50yrs

Report of Patients
under 5

Font Style:- Bahnschrift

Font Size:- 36, 28, 8

Font Colour:- White

Report 1 Alternative design

Description & Purpose of Report 1-

This will be a report of DE1 Patients that have moved to Health Centre, its purpose is to show a neat list of DE1 patients that need to be informed that they have been transferred to the Health Centre, which would contain their name, address, town and postcode, which would be sorted alphabetically by postcode.

Design



DE1 Patients moved to Health Centre

PLACEHOLDER

First Name	Last Name	Address	Town	Postcode

Font Style:- Bahnschrift

Font Size:- 28, 14

Font Colour:- White

Report 2 Alternative design

Description & Purpose of Report 2 -

This will be a report that displays all DE1 patients that have missed appointments within the last month, it would display First Name, Last Name, contact number, and appointment date, sorted alphabetically by surname.

Design



DE1 Patients missed appts in last month

PLACEHOLDER

First Name	Last Name	Contact Number	Appointment Date

Font Style:- Bahnschrift

Font Size:- 26, 14

Font Colour:- White

Report 3 Alternative design

Description & Purpose of Report 3-

This will be a report that lists all new patients within the last six months so they can easily visualize and share the growth of the surgery, so they can monitor whether they need more doctors, it would need to list First Name, Last Name, Date Registered, and their Doctors Name, and be sorted in descending order by date.

Design



New patients in last six months

PLACEHOLDER

First Name	Last Name	Date Registered	Doctors Name

Font Style:- Bahnschrift

Font Size:- 28, 14

Font Colour:- White

SECTION 7: JUSTIFICATION

The purpose of this database for the surgery is to help improve efficiency and help convince the rest of the organization to convert from their paper-based system to our digital database system, as there is a planned reorganization of health services in the area, which the paper-based system would reduce efficiency as more patients and doctors arrive.

For the design of my database system, the objects I've chosen effectively meet the requirements of the surgery. The tables of patient and appointment data are relational, which links them together for easy expansion and integration of their data, which will be essential for growth of the surgery and organization, as it can infinitely grow. I created simple and easy to use forms for both the patient and appointment tables, which meet their requirements, as they have been designed to make it easy to manage data without it being overwhelming, so they won't have to manually input into the table. I also created interfaces for navigating forms, reports, and queries, so the user doesn't have to interact with the overwhelming menus within Access. I created queries as the requirements described for the surgery, to ensure that the surgery can quickly access their required data in their defined parameters whenever needed, which I then created designed reports that display this data in a neat, printable sheet to share amongst the staff within the surgery. I added navigation buttons around the menus to keep it simple and quick for the user, they don't have to touch the menus in Access and they can get to what they want within a few clicks, and easily manage data without having to again, interact outside the designed menus.

My choices of the designs of the menus and forms I believe will be helpful for the surgery in many ways, the colors I chose fit the medical theme, as it matches the colors of the NHS, keeping it professional. I've kept the layout simple, and designed in a common format, this is so less time is spent during training understanding the layout and keeping it uncluttered, so less time is wasted looking at unnecessary data. In both designs I chose fonts that aren't flashy, in the initial design I chose a generic font that is commonly used, and wouldn't seem out of place in a professional setting, in the alternative design I chose a more uncommon font but kept to the professional style while having it be more pleasant to look at, instead of just a boring generic outdated font. I made the text size large to ensure staff and or patients with poor eyesight can easily read the data displayed on screen, this will ensure less problems arise as the database system gets used further.

The navigation within the database will greatly improve productivity, as the buttons and structure within forms make it easy to add and navigate through data, instead of manually searching, editing, adding, etc. Compared to paper-based, the productivity would be much greater, as less time is wasted, and it becomes a less tedious task.

The queries are designed to the requirements as described, I carefully matched the criteria that the queries wanted, sorted the data by how the queries asked, displayed that data that needed to be showed, which all-in return makes the queries output the desired data as matched with the

ones described within the query section of the requirements. The queries will be helpful for the surgery, as it will help them easily analyze the data have, as the digital database system will be able to quickly grab all data that matches with the query we defined almost instantly, which would massively cut out wasted time searching for data in a paper-based system, which could have human errors, and take possibly hours or even days to fully collect.

The reports will help management make decisions as it gives them an easy way to view all the data they need, which could be printed and given out to others, and being able to visualize the data in person still gives the option to use paper if needed, in cases like mailing to other upper management, simply wanting to see the amounts of data in person, or keeping physical copies to ensure data integrity or in emergencies.

The validation within forms will ensure that the data entered will be accurate, paper-based would allow you to make mistakes, and could easily get stored if there are no additional checks by other staff. The digital database system will validate the data for the user, if they were to input for example an invalid email, it would notify the user and prevent it from being stored, keeping the data accurate.

In conclusion the decisions I have chosen above, I believe meet the purpose and requirements of the health authority and surgery, because I have included their requirements as much as possible into the designs, and taken into consideration of the limited resources we have, and that we must make a good impression to convince the other surgeries in the health authority to move forward with their switch to our digital database system. The database I have created will help the health authority greatly, as it greatly reduces wasted time sifting through thousands of paper documents, even for just one patient, and it will help them expand with the reorganization, as it would make it easier to manage the ever-growing amounts of patients of doctors, as the paper-based system isn't a viable option for such a critical service, with it being easy to destroy, easy to steal, taking up large unnecessary amounts of space, which could be used by patients and doctors that need it.